Rajiv V. Joshi

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Education

Carnegie Mellon University Pittsburgh, PA
Master of Science in Mechanical Engineering - Research | GPA: 4.0/4.0 May 2026

Virginia Tech Blacksburg, VA
Bachelors of Science in Mechanical Engineering - Robotics and Mechatronics | GPA: 3.64/4.0 May 2024
Bachelors of Science in Mechanical Engineering - Mechanical | GPA: 3.64/4.0 May 2024

Coursework

Masters: ADV Mechatronics (TA) | Haptic Design Interfaces | Modern Control Theory | Computer Vision for Engineers (In-Progress) | ML and AI for Engineers (In-Progress)

Undergraduate: Robotics and Automation | Kinematics | Differential Equations | Deforms | Dynamics

Skills

Embedded Systems: Arduino | STM 32 | GPU Based RTOS | Oscilloscope | DMM | Serial Communication (SPI, I2C, CAN)

Manufacturing: PCB Design | Soldering | Circuit Schematics | Rapid Prototyping | 3-D printing | Machining

Software/Programming: Python | C/C++ | MATLAB | SolidWorks | Fusion 360 | KiCAD | ANSYS | Linux Command Line

Learning: ROS/ROS 2 (Exoskeleton Framework) | OpenCV (CV class project) | Human-in-the-loop optimization (Research)

Research Experience

MetaMobility Lab at Carnegie Mellon University Pittsburgh, PA
Knee Exoskeleton – MS Research Project April 2025 - Present

- Coordinated and led a team of 5 students in research, design, and manufacturing a lightweight knee exoskeleton capable of delivering 20 Nm of torque directly to knee joints for people suffering from osteoarthritis or gait impairment
- Designed compact PCB implementing SPI protocol, in C/C++, with IMU sensors through CAT6 connectors and twisted pair cables to shield signal from noise, improving sensor polling frequency from 100Hz to 500Hz and reducing latency
- Analyzing trends of knee kinematics and kinetics, using open-source CAMARGO dataset, to implement human-in-the-loop optimization and Delayed Output Feedback Control, creating optimal knee assistance profiles during various tasks
- Applying ROS/ROS2 framework to allow live exo-controller adaptation and robust communication between a Jetson Orin Nano GPU and Teensy 4.1 MCU to log data from multiple onboard FSRs, IMUs, and AK80-9 actuators

MetaMobility Lab at Carnegie Mellon University Pittsburgh, PA
Hip Exoskeleton – MS Research May 2024 - Present

- Optimized hip exoskeleton hardware for improved fit to users referencing human biomechanics, design in SolidWorks, and rapid prototyping with 3D printing, yielding comfortable motor-joint alignment and 25% reduced system weight
- Redesigned PCB schematic, in KiCAD, to integrate CAN bus and I2C protocols, in Python and C/C++, with embedded system peripherals, reducing connections count from 8 to 4 while maintaining signal integrity and minimized latency
- Debugged multiple hardware and firmware faults using an Oscilloscope and DMM to visualize logic signals, voltage values, and circuit continuity to rapidly diagnose causes and iterate/manufacture immediate solutions

Academic Projects

Carnegie Mellon University Pittsburgh, PA
Haptic Design Interfaces: 2 DoF Haptic Hand Interface – Course Project April 2025 - May 2025

- Developed an open-source compact haptic fingers device, with a team of 6 graduate students, meshing STM32 MCU, 4 low torque motors, and 3D-printed parts as a teaching tool for robot kinematics, biological sensors, and haptics
- Visualized a finger joint workspace in MATLAB to define mechanical operational ranges of a capstan-driven end-effector, controlled in C/C++ by applying PID motor control and forward kinematics to render virtual 3D objects

Carnegie Mellon University Pittsburgh, PA
ADV Mechatronics: Embedded System Game Design – Course Project November 2024 - December 2024

- Implemented PWM, timer interrupts, and event driven state-machine on an STM32 for controlling a custom H-bridge motor driver circuit, in whack-a-mole style arcade game designed with a team of 3
- Lead creation of detailed circuit schematics, using KiCAD, for guiding protoboard soldering/manufacturing, optimizing cable routing between components, and assisting debugging circuit/logic issues with an oscilloscope and DMM